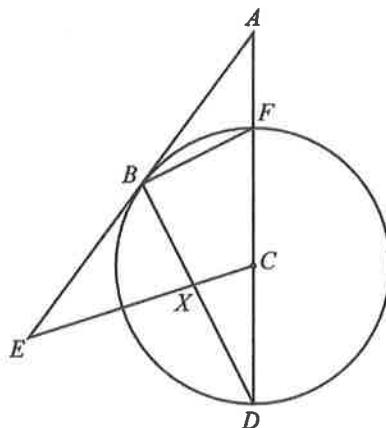


Answer **all** the questions.

1. The coefficient of x^3 in the expansion of $(k + 2x)\left(2 - \frac{1}{2}x\right)^6$ is zero.
Find the value of the constant k .

[5]

2.



The diagram shows a circle, centre C , with FD as diameter. The tangent at a point B on the circle meets DF extended at the point A . Point E lies on AB extended and X is the point of intersection of BD with EC .

(a) Prove that triangle ABD is similar to triangle AFB .

[4]

A circle can be drawn through B , C and X with BC as diameter.

(b) What type of quadrilateral is $EBFC$? Give reasons to support your answer.

[5]

3. Tea is poured into an empty cup. The temperature, T_c °C, of the tea in the cup, t minutes after it is poured, is modelled by the formula $T_c = 86e^{-0.06t}$.

(a) State the initial temperature of the tea.

[1]

(b) Find the time taken for the temperature of the tea to drop to 37 °C.

[3]

- (c) Some tea is poured into an empty cup and at the same time the same volume of tea is poured into an empty flask. The temperature, T_f °C, of the tea in the flask at time t minutes after it is poured into the flask is modelled by $T_f = 86e^{-\lambda t}$ where λ is a constant. The formula for T_c still applies.

(i) After one hour the temperature of the tea in the flask is 82 °C. Find λ .

[2]

- (ii) Using your answer from **part (c)(i)** find the time when the temperature of the tea in the cup is half the temperature of the tea in the flask. [3]

4. Do not use a calculator in this question.

(a) Use the identity for $\tan 2A$ to show that $\tan 15^\circ = 2 - \sqrt{3}$.

[5]

(b) Prove that $\tan 15^\circ - \tan 105^\circ = 4$.

[5]

5. When $y = 3 \cos 3x - 5 \sin 3x$, the expression $p \frac{d^2y}{dx^2} + q \frac{dy}{dx} + 14y + 34 \sin 3x$ may be written in the form $A \cos 3x + B \sin 3x$ where p, q, A and B are constants.

(a) Show that $A = 42 - 27p - 15q$ and find B in terms of p and q .

[7]

(b) In the case when A and B are both zero, find the values of p and q .

[3]

6. (a) Show that $2x + 1$ is a factor of $6x^3 - 7x^2 - x + 2$ and hence factorise $6x^3 - 7x^2 - x + 2$ completely.

[5]

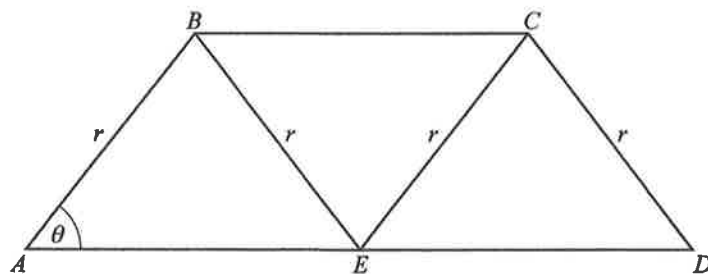
(b) Solve the equation $6(4^y) + 2(2^{-y}) = 7(2^y) + 1$.

[5]

(c) Show that the negative solution from **part (b)** may be written in the form $1 - \log_a b$ where a and b are integers to be determined. [2]

7. (a) The graph of $y = a \cos bx$ has one maximum point at $\left(\frac{7\pi}{2}, 10\right)$ and the next maximum point after this has coordinates $\left(\frac{9\pi}{2}, 10\right)$. Find the values of the constants a and b . [2]
- (b) A particle, travelling in a straight line, has velocity, v m/s, at time t seconds, $t \geq 0$, given by $v = 5 \sin 0.2t + c$, where c is a positive constant.
- (i) Find the smallest value of c in order that the particle will never change its direction. [3]
- (ii) In the case where $c = 8$ find the total distance travelled by the particle in the third second of its motion. [6]

8.



The diagram shows a symmetrical wooden framework $ABCDEA$ consisting of seven pieces of wood. AED is a straight line. The sections AB , BE , EC and CD are each of length r m, where r is a constant. Angle $BAE = \theta$ for $0^\circ < \theta < 90^\circ$.

- (a) Express the area enclosed by the framework in terms of r and $\sin 2\theta$. [3]

- (b) Given that θ can vary, find, in terms of r , the maximum possible area enclosed by the framework and the value of θ at which this occurs. You are **not** required to justify that this area is a maximum. [2]

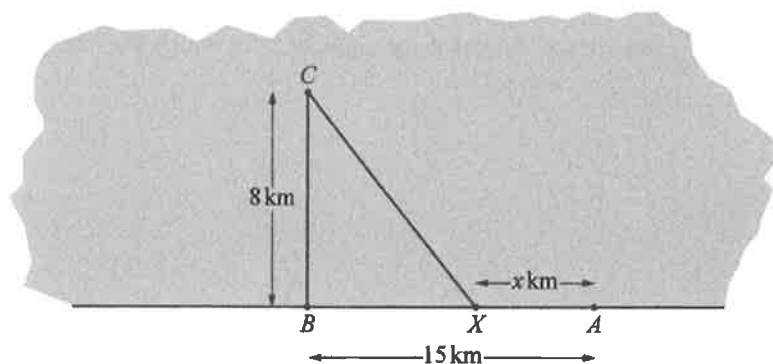
A similar framework is made from a single piece of wood of length $8r \sin \theta$ m. There is no wood wasted.

(c) Show that $4 \sin \theta - 3 \cos \theta = 2$.

[2]

(d) By expressing $4 \sin \theta - 3 \cos \theta$ in the form $R \sin (\theta - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$, find the value of θ . [5]

9.



The diagram shows rough level ground (shaded) bordered by a straight level road BXA . Point C is on the rough ground and B is the point on the road closest to C . The distance AB is 15 km and BC is 8 km.

A cross-country runner is at point A on the road and needs to get to point C as quickly as possible. The runner can maintain a speed of 5 km/h on the road but only 3 km/h over the rough ground.

The runner realises that in order to minimise the total time to get to C , it is necessary to leave the road at some point X between A and B and then run directly from X to C .

Let AX be x km and T be the total time, in hours, taken by the runner to get from A to X and then from X to C .

(a) Show that $T = \frac{x}{5} + \frac{\sqrt{x^2 - 30x + 289}}{3}$.

[3]

- (b) Find an expression for $\frac{dT}{dx}$ and hence show that the value of x for which the time taken is a minimum satisfies the equation $x^2 - 30x + 189 = 0$. [6]

- (c) Hence find the minimum time taken, in hours and minutes, for the runner to get from A to C . You are **not** required to justify that this time is a minimum. [3]